Aryan Patel

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Education

Georgia Institute of Technology Atlanta, GA

Bachelor of Science in Computer Science - GPA 3.94/4.00 *December 2027*

- Coursework: Data Structures & Algorithms, Design & Analysis, Computer Organization & Programming, Systems & Networks, Intro to AI, Linear Algebra, Object-Oriented Programming, Objects and Design

Experience

Capital One June 2026 – August 2026

Software Engineer Intern *McLean, VA*

- Cut technical search time 60% for 250+ engineers by engineering a LangChain hybrid retrieval pipeline over 500+ internal engineering documents, pairing semantic embeddings that match by meaning with exact keyword search
- Saved 50+ engineer-hours monthly and cut telemetry diagnosis time 32% by developing an observability agent that pulls and analyzes New Relic and Splunk telemetry in one place, helping engineers pinpoint the source of issues
- Raised evaluated answer quality 38% by designing a LangGraph multi-agent assistant that routes each engineering question to the specialized agent best suited to it and shares retrieved context across agents and workflows

Georgia Institute of Technology August 2025 – Present

Undergraduate Researcher for Autonomous and Connected Transportation Driving Simulator *Atlanta, GA*

- Supported a 1,000-participant field study by developing Swift iOS features for GPS-based mobility tracking
- Cut analysis time 30% by building a Grafana dashboard that monitors participant activity and mobility trends
- Improved data reliability for travel-behavior analysis by processing GPS telemetry and flagging data-quality issues

Molecular Evolution Core Lab March 2025 – December 2025

Software Engineer Intern *Atlanta, GA*

- Centralized research and service information for 200+ clients by building the lab's full-stack MERN website
- Cut equipment rental processing time 80% for 23 lab staff by automating rental tracking in MongoDB
- Streamlined client communication for lab staff by developing a responsive contact system that centralizes inquiries

Projects

Recall | *Rust, Multithreading, Performance Benchmarking*

- Built a Rust search engine from scratch with an inverted index, BM25 top-K ranking, and multi-term queries
- Cut heavy-query latency 3× and sped up indexing up to 2.5× by parallelizing indexing and search across 4 shards
- Cut skewed-query latency 6.9× and peak memory 58% by adding adaptive galloping and posting-list compression

AWS Podverse | *AWS Lambda, S3, DynamoDB, Cognito, EC2, Next.js, React*

- Cut latency 50% by building a serverless podcast platform on auto-scaling AWS Lambda with a Next.js front end
- Boosted scalability and stability 60% by managing podcast metadata and users in DynamoDB, with audio in S3
- Ensured 99.9% uptime by hosting the platform on AWS EC2 and integrating Cognito for secure sign-up and login

AI Study Tool for Pharmacy Students | *JavaScript, React, Node.js, Express.js, MongoDB, OpenAI API*

- Developed Pharma, a GPT-4o study tool with a pharmacy Q&A chatbot and personalized flashcard generation
- Built an Express.js REST API on MongoDB that stores and serves user data and GPT-4o-generated flashcards
- Secured user logins and sessions by implementing JWT authentication with token refresh and expiry handling

Technical Skills

Languages: Python, Rust, Go, Java, C/C++, TypeScript, JavaScript, Swift, SQL
Systems: Multithreading, Concurrency, Memory Management, Performance Benchmarking
AI/LLM: LangChain, LangGraph, Hybrid Retrieval (Embeddings + Keyword Search), OpenAI API
Cloud & DevOps: AWS (Lambda, S3, DynamoDB, EC2, Cognito), Docker, Kubernetes, CI/CD, Git
Observability & Testing: New Relic, Splunk, Grafana, JMeter
Frameworks & Databases: React, Next.js, Node.js, Express.js, Django, PostgreSQL, MongoDB, SQLite